

Instituto Tecnológico de Las Américas

ITLA

Laboratorio: Jump Server con VPN de acceso remoto

Práctica 3 – Infraestructura 2: RemoteApp, RD Web Client y políticas por usuario

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LINK DEL VIDEO: [Tarea Semana 4 \(Practica 3\)](#)

1. Propósito del laboratorio

En esta infraestructura los usuarios no llegan directo a los servidores. Primero se conectan por una VPN de acceso remoto al FortiGate, y esa VPN solo les da acceso al Jump Server. Desde el Jump Server, con RemoteApps publicadas en el RD Web Client, trabajan con el Web Server, al que el Jump solo puede llegar por HTTPS, RDP y SSH.

Se crearon dos usuarios: Carlos Pérez (cperez), sin privilegios, que solo ve la web del Sistema de Caja y tiene el SSH bloqueado por una política explícita, y María Rodríguez (mrodriguez), con privilegios, que ve la web, PuTTY y el Escritorio Remoto. La configuración del FortiGate se hizo por la GUI.

2. Topología y direccionamiento

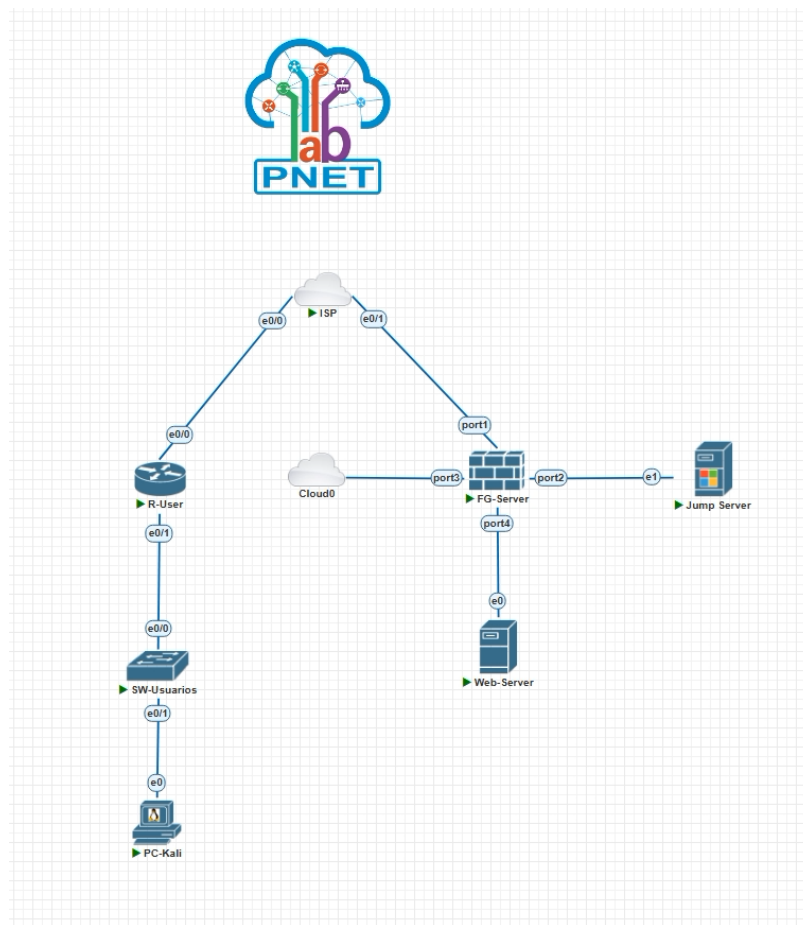


Figura 1. Topología implementada en PNETLab.

Segmento	Red	Gateway	Equipos
WAN R-USER	20.24.23.0/30	20.24.23.1 (ISP)	R-USER .2
WAN FortiGate	20.24.56.0/30	20.24.56.1 (ISP)	FG-Server port1 .2
VLAN 10 Usuarios	10.23.56.0/25	10.23.56.1 (R-USER)	DHCP .10 a .100
LAN-WEB (port4)	10.23.56.128/29	10.23.56.129	Web-Server .130
LAN-JUMP (port2)	10.23.56.136/29	10.23.56.137	Jump Server .138
Pool VPN	10.23.56.200 a .210	—	Una IP /32 por cliente

3. FortiGate

3.1 Interfaces

The screenshot shows the 'Edit Interface' configuration for 'port4' (LAN-WEB). The configuration includes the following details:

- Name:** port4
- Alias:** LAN-WEB
- Type:** Physical Interface
- Role:** LAN
- Addressing mode:** Manual (selected), DHCP, One-Arm Sniffer
- IP/Netmask:** 10.23.56.129/255.255.255.248
- Create address object matching subnet:** Enabled
- Name:** port4 address
- Destination:** 10.23.56.129/255.255.255.248
- Secondary IP address:** Disabled
- Administrative access:**
 - IPv4: ☐ HTTPS, ☒ PING, ☐ FMG-Access, ☐ SSH, ☐ SNMP, ☐ FTM, ☐ RADIUS Accounting, ☐ Security Fabric Connection
 - Receive LLDP: ☒ Use VDOM Setting, ☒ Enable, ☐ Disable
 - Transmit LLDP: ☒ Use VDOM Setting, ☒ Enable, ☐ Disable
- DHCP Server:** Disabled
- Network:**
 - Device detection: Enabled
 - Security mode: Disabled
- Traffic Shaping:**
 - Outbound shaping profile: Disabled
- Miscellaneous:**
 - Comments: (empty)

Buttons at the bottom: OK, Cancel

Figura 2. Interfaz port4 (LAN-WEB) con la IP 10.23.56.129/29.

Edit Interface

Name: port2
 Alias: LAN-JUMP
 Type: Physical Interface
 Role: LAN

Address

Addressing mode: Manual DHCP One-Arm Sniffer
 IP/Netmask: 10.23.56.137/255.255.255.248
 Create address object matching subnet: ☐
 Secondary IP address: ☐

Administrative access

IPv4: ☐ HTTPS ☒ PING ☐ FMG-Access
☐ SSH ☐ SNMP ☐ FTM
☐ RADIUS Accounting ☐ Security Fabric Connection
 Receive LLDP: Use VDOM Setting Enable Disable
 Transmit LLDP: Use VDOM Setting Enable Disable

☐ DHCP Server

Network

Device detection: ☒
 Security mode: ☐

Traffic Shaping

Outbound shaping profile: ☐

Miscellaneous

Comments: 0/255
 Status: ☒ Enabled ☐ Disabled

OK Cancel

Figura 3. Interfaz port2 (LAN-JUMP) con la IP 10.23.56.137/29.

```

CLI Console
FG-Server # execute ping 10.23.56.130
command parse error before 'ping'
Command fail. Return code -61

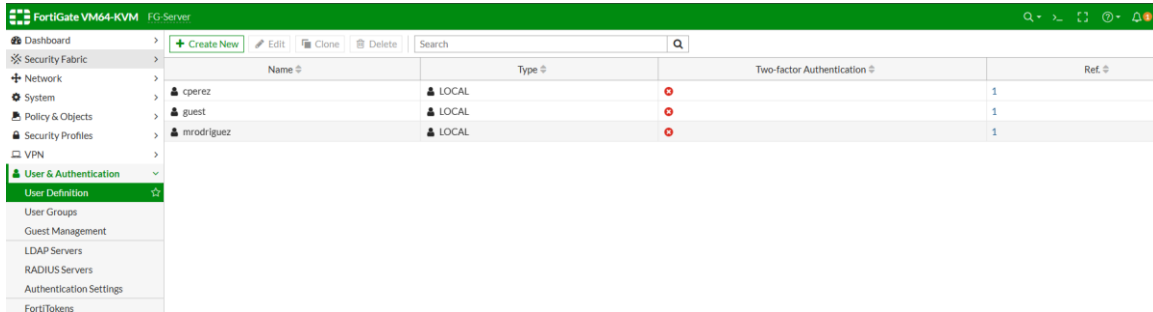
FG-Server # execute ping 10.23.56.130
PING 10.23.56.130 (10.23.56.130): 56 data bytes
64 bytes from 10.23.56.130: icmp_seq=0 ttl=64 time=0.6 ms
64 bytes from 10.23.56.130: icmp_seq=1 ttl=64 time=0.4 ms
64 bytes from 10.23.56.130: icmp_seq=2 ttl=64 time=0.6 ms
64 bytes from 10.23.56.130: icmp_seq=3 ttl=64 time=0.5 ms
64 bytes from 10.23.56.130: icmp_seq=4 ttl=64 time=0.4 ms

--- 10.23.56.130 ping statistics ---
5 packets transmitted, 5 packets received, 0% packet loss
round-trip min/avg/max = 0.4/0.5/0.6 ms

FG-Server #
  
```

Figura 4. El FortiGate llega al Web Server por el port4.

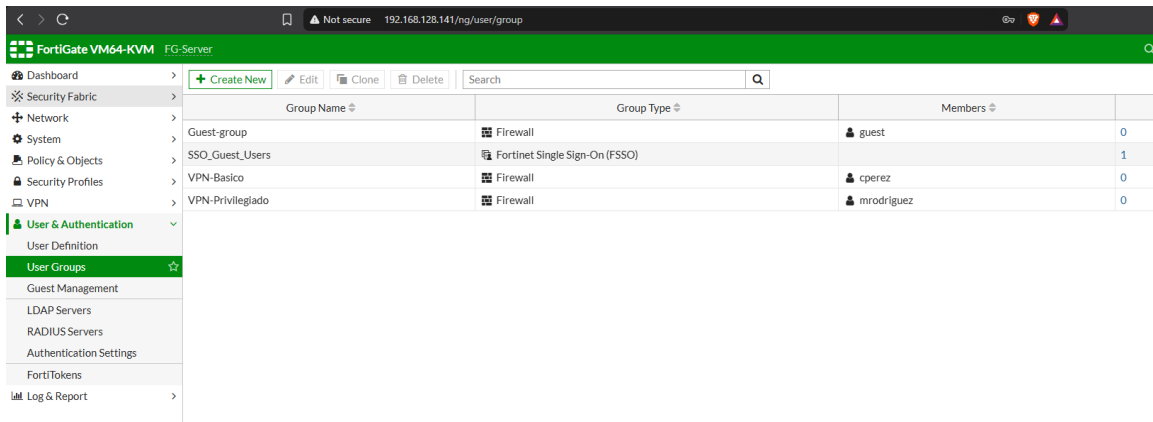
3.2 Usuarios y grupos de la VPN



The screenshot shows the FortiGate VM64-KVM User Definition page. The left sidebar has 'User & Authentication' selected, with 'User Definition' highlighted. The main area displays a table of local users.

Name	Type	Two-factor Authentication	Ref
cperez	LOCAL		1
guest	LOCAL		1
mrodriguez	LOCAL		1

Figura 5. Usuarios locales cperez y mrodriguez.



The screenshot shows the FortiGate VM64-KVM User Groups page. The left sidebar has 'User & Authentication' selected, with 'User Groups' highlighted. The main area displays a table of user groups.

Group Name	Group Type	Members
Guest-group	Firewall	0
SSO_Guest_Users	Fortinet Single Sign-On (FSSO)	1
VPN-Basico	Firewall	0
VPN-Privilegiado	Firewall	0

Figura 6. Grupos VPN-Basico y VPN-Privilegiado.

3.3 VPN de acceso remoto

IKEv1 en modo agresivo con PSK y XAuth, DES/MD5 y DH 5 por la licencia de evaluación. El grupo de XAuth se hereda de la política y el split tunnel solo manda la ruta del Jump Server.

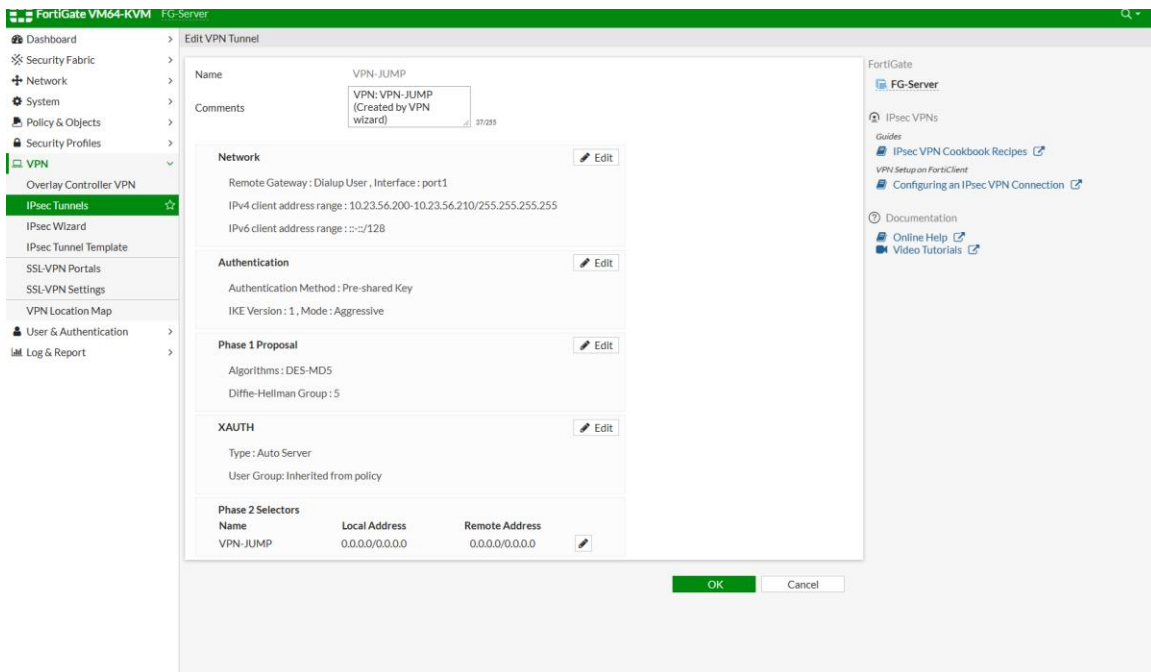


Figura 7. Túnel VPN-JUMP convertido a personalizado.

Phase 2 Selectors

Name	Local Address	Remote Address
VPN-JUMP	0.0.0.0/0.0.0.0	0.0.0.0/0.0.0.0

Edit Phase 2

Name: VPN-JUMP

Comments: VPN: VPN-JUMP (Created by VPN wizard)

Local Address: Subnet 0.0.0.0/0.0.0.0

Remote Address: Subnet 0.0.0.0/0.0.0.0

Advanced...

Phase 2 Proposal Add

Encryption: DES Authentication: MD5

Enable Replay Detection ☒

Enable Perfect Forward Secrecy (PFS) ☒

Diffie-Hellman Group: ☐ 32 ☐ 31 ☐ 30 ☐ 29 ☐ 28 ☐ 27 ☐ 21 ☐ 20 ☐ 19 ☐ 18 ☐ 17 ☐ 16 ☒ 5 ☐ 2 ☐ 1

Local Port: All ☒

Remote Port: All ☒

Protocol: All ☒

Autokey Keep Alive: ☒

Key Lifetime: Seconds

Seconds: 43200

Figura 8. Phase 2 con DES/MD5, PFS y grupo DH 5.

3.4 Políticas

FortiGate VM64-KVM FG-Server											admin		
Create New Edit Delete Policy Lookup Search											Interface Pair View		By Sequence

Figura 9. Políticas configuradas en el FortiGate (4 de 5).

Política	Origen → Destino	Servicio	Acción
Deny-SSH-Basico	VPN-Basico → Jump-Server	SSH	DENY con log
VPN-Basico-a-Jump	VPN-Basico → Jump-Server	HTTPS	ACCEPT
VPN-Priv-a-Jump	VPN-Privilegiado → Jump-Server	HTTPS, RDP	ACCEPT
Jump-a-Web	Jump-Server → Web-Server	HTTPS, RDP, SSH	ACCEPT
Implicit Deny	any → any	ALL	DENY con log

4. Jump Server

Windows Server 2022 con el dominio itla.local, Remote Desktop Services, licencias por usuario, RD Web Client y RD Gateway.

```
PS C:\Users\Administrator> hostname
itla.local
PS C:\Users\Administrator> ipconfig
>> ping 10.23.56.137
YOUNG-SERVER

Windows IP Configuration

Ethernet adapter INTERNET-TEMP:

    Connection-specific DNS Suffix  . : localdomain
    Link-local IPv6 Address . . . . . : fe80::1ab0:9651:6a37:dc3%5
    IPv4 Address. . . . . : 192.168.128.155
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.128.2

Ethernet adapter LAN-JUMP:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::965c:1c2d:946a:8484%3
    IPv4 Address. . . . . : 10.23.56.138
    Subnet Mask . . . . . : 255.255.255.248
    Default Gateway . . . . . : 

Pinging 10.23.56.137 with 32 bytes of data:
Reply from 10.23.56.137: bytes=32 time<1ms TTL=255
Reply from 10.23.56.137: bytes=32 time<1ms TTL=255
Reply from 10.23.56.137: bytes=32 time<1ms TTL=255
Reply from 10.23.56.137: bytes=32 time<1ms TTL=255

Ping statistics for 10.23.56.137:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
PS C:\Users\Administrator>
```

Figura 10. Jump Server con la IP 10.23.56.138/29 en la tarjeta LAN-JUMP.

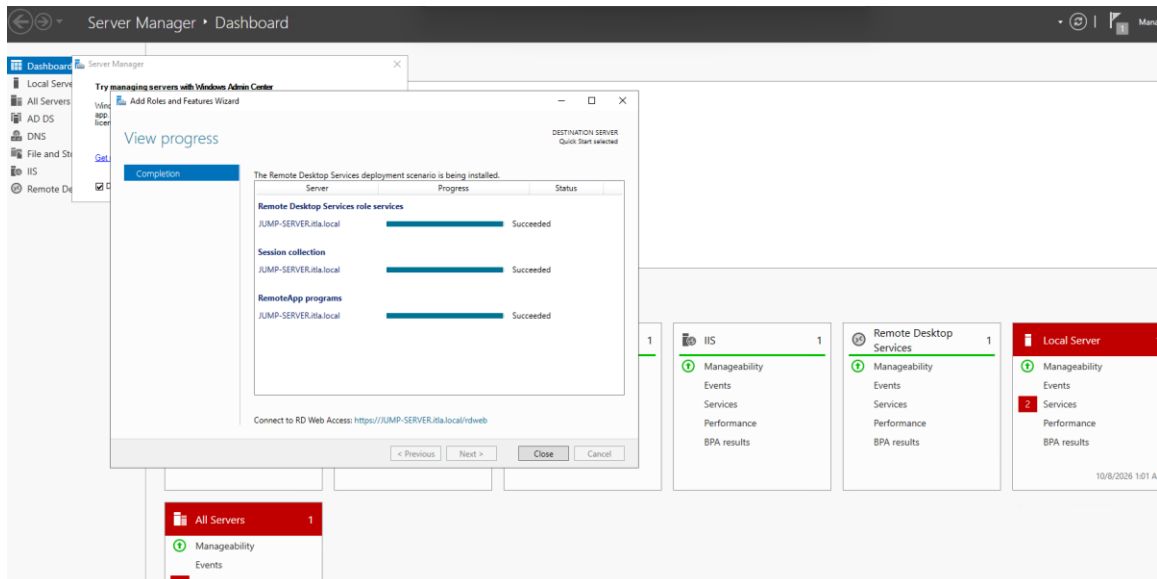


Figura 11. Despliegue de Remote Desktop Services terminado.

```

PS C:\Users\Administrator> Get-RDRemoteApp -CollectionName "QuickSessionCollection" -ConnectionBroker $cb | ft DisplayName, UserGroups
>> Get-RDSessionCollectionConfiguration -CollectionName "QuickSessionCollection" -UserGroup -ConnectionBroker $cb

DisplayName      UserGroups
-----
Sistema de Caja (Web) {ITLA\RDS-Basico, ITLA\RDS-Privilegiado}
PuTTY (SSH)       {ITLA\RDS-Privilegiado}
Escritorio Remoto (RDP) {ITLA\RDS-Privilegiado}

CollectionName    UserGroup
-----
QuickSessionCollection {ITLA\RDS-Basico, ITLA\RDS-Privilegiado}

PS C:\Users\Administrator> Add-RDServer -Server $cb -Role RDS-LICENSING -ConnectionBroker $cb
>> Set-RDLicenseConfiguration -LicenseServer $cb -Mode PerUser -ConnectionBroker $cb -Force
>> # Verificar
>> Get-RDLicenseConfiguration -ConnectionBroker $cb

Server              Roles
-----
JUMP-SERVER.itla.local {RDS-LICENSING}

Mode                : PerUser
LicenseServer       : {JUMP-SERVER.itla.local}

PS C:\Users\Administrator>

```

Figura 12. RemoteApps por grupo y licencias en modo PerUser.

```

PS C:\Users\Administrator> Import-Module PowerShellGet -MinimumVersion 2.2.5
>> Get-Module PowerShellGet | ft Name,Version

Name      Version
-----
PowerShellGet 2.2.5

PS C:\Users\Administrator> [Net.ServicePointManager]::SecurityProtocol = [Net.SecurityProtocolType]::Tls12
>> Install-Module -Name RDWebClientManagement -Force -AcceptLicense
>> Install-RDWebClientPackage
>> Import-RDWebClientBrokerCert "C:\Certs\jump-server.cer"
>> Publish-RDWebClientPackage -Type Production -Latest
>> Get-RDWebClientPackage
WARNING: Initializing RDWebClientManagement in 'C:\Program Files\RemoteDesktopWeb'. To uninstall, use
Uninstall-RDWebClient.
WARNING: Using the Remote Desktop web client with per-device licensing is not supported.

packageId      : rd-html5
version        : 2.1.85.0
author         : Microsoft
summary        : The Remote Desktop Web Client
minRDWebClientManagementVersion : 1.0.4
path           : C:\Program Files\RemoteDesktopWeb\Internal\Clients\Sylb0wyt.15j
_baseVersion   : 2.1.85.0
publishedAs    : {Production}

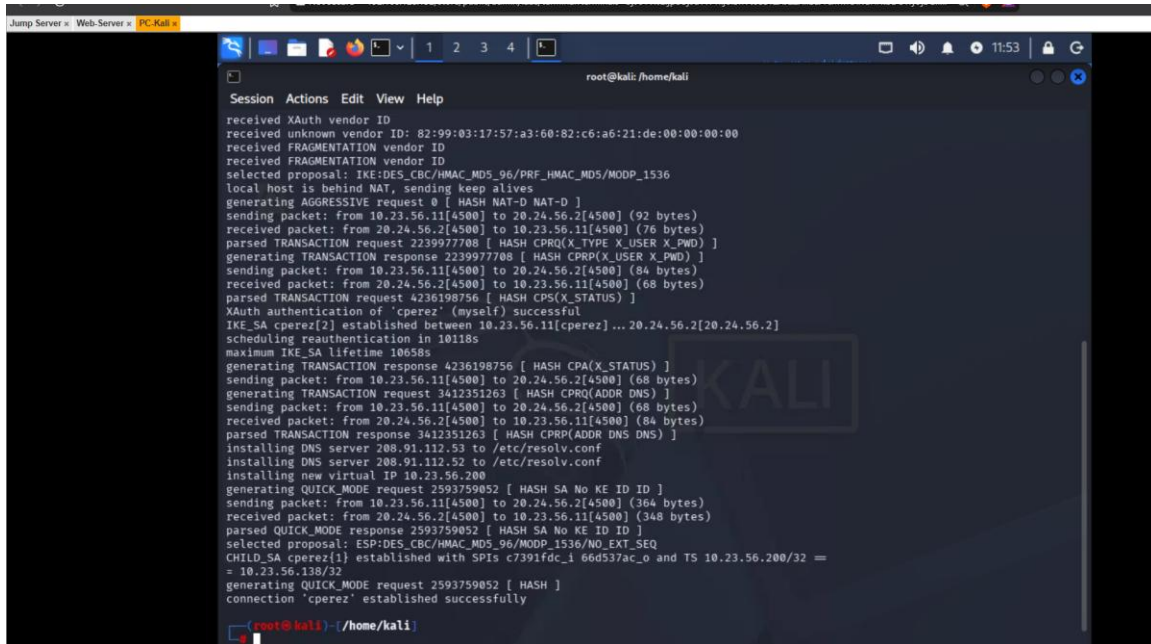
PS C:\Users\Administrator>

```

Figura 13. RD Web Client instalado y publicado.

5. Pruebas

5.1 Usuario sin privilegios (cperez)

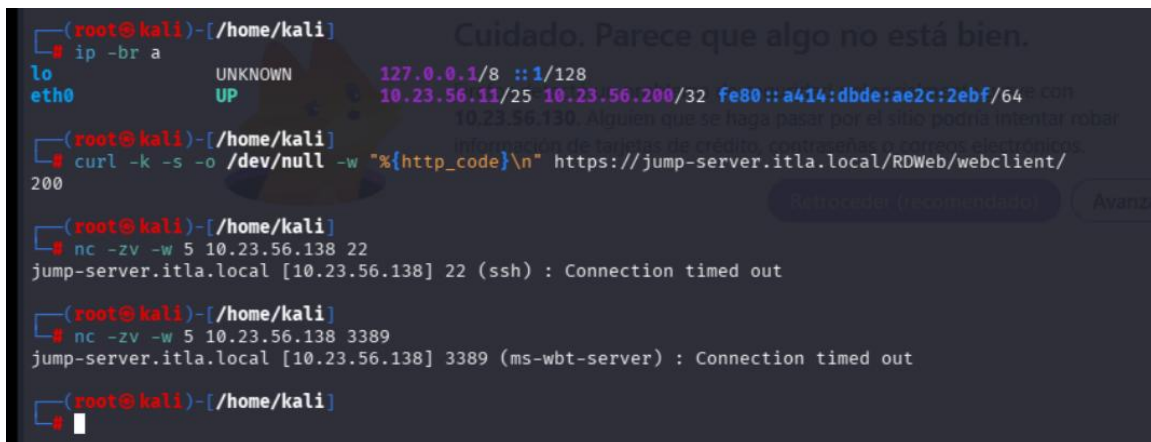


```
root@kali: /home/kali
Session Actions Edit View Help

received XAuth vendor ID
received unknown vendor ID: 82:99:03:17:57:a3:60:82:c6:a6:21:de:00:00:00
received FRAGMENTATION vendor ID
received FRAGMENTATION vendor ID
selected proposal: IKE:DES_CBC/HMAC_MD5_96/PRF_HMAC_MD5/MODP_1536
local host is behind NAT, sending keep alives
generating AGGRESSIVE request 0 [ HASH NAT-D NAT-D ]
sending packet: from 10.23.56.11[4500] to 20.24.56.2[4500] (92 bytes)
received packet: from 20.24.56.2[4500] to 10.23.56.11[4500] (76 bytes)
parsed TRANSACTION request 2239977708 [ HASH CPRQ(X_TYPE X_USER X_PWD) ]
generating TRANSACTION response 2239977708 [ HASH CPRP(X_USER X_PWD) ]
sending packet: from 10.23.56.11[4500] to 20.24.56.2[4500] (84 bytes)
received packet: from 20.24.56.2[4500] to 10.23.56.11[4500] (68 bytes)
parsed TRANSACTION request 4236198756 [ HASH CPS(X_STATUS) ]
XAuth authentication of 'cperez' (myself) successful
IKE_SA cperez[2] established between 10.23.56.11[cperez] ... 20.24.56.2[20.24.56.2]
scheduling reauthentication in 10118s
maximum IKE_SA lifetime 10658s
generating TRANSACTION response 4236198756 [ HASH CPA(X_STATUS) ]
sending packet: from 10.23.56.11[4500] to 20.24.56.2[4500] (68 bytes)
generating TRANSACTION request 3412351263 [ HASH CPRQ(ADDR DNS) ]
sending packet: from 10.23.56.11[4500] to 20.24.56.2[4500] (68 bytes)
received packet: from 20.24.56.2[4500] to 10.23.56.11[4500] (84 bytes)
parsed TRANSACTION response 3412351263 [ HASH CPRP(ADDR DNS) ]
installing DNS server 208.91.112.53 to /etc/resolv.conf
installing DNS server 208.91.112.52 to /etc/resolv.conf
installing new virtual IP 10.23.56.200
generating QUICK_MODE request 2593759052 [ HASH SA No KE ID ID ]
sending packet: from 10.23.56.11[4500] to 20.24.56.2[4500] (364 bytes)
received packet: from 20.24.56.2[4500] to 10.23.56.11[4500] (348 bytes)
parsed QUICK_MODE response 2593759052 [ HASH SA No KE ID ID ]
selected proposal: ESP:DES_CBC/HMAC_MD5_96/MODP_1536/NO_EXT_SEQ
CHILD_SA cperez[1] established with SPIs c7391fdc.1 66d537ac.0 and TS 10.23.56.200/32 ==
= 10.23.56.138/32
generating QUICK_MODE request 2593759052 [ HASH ]
connection 'cperez' established successfully

root@kali) ~ (/home/kali)
```

Figura 14. cperez conecta la VPN y recibe la IP 10.23.56.200.



```
(root@kali)~[/home/kali]
# ip -br a
lo        UNKNOWN    127.0.0.1/8 ::1/128
eth0      UP            10.23.56.11/25 10.23.56.200/32 fe80::a414:dbde:ae2c:2ebf/64

(root@kali)~[/home/kali]
# curl -k -s -o /dev/null -w "%{http_code}\n" https://jump-server.itla.local/RDWeb/webclient/
200

(root@kali)~[/home/kali]
# nc -zv -w 5 10.23.56.138 22
jump-server.itla.local [10.23.56.138] 22 (ssh) : Connection timed out

(root@kali)~[/home/kali]
# nc -zv -w 5 10.23.56.138 3389
jump-server.itla.local [10.23.56.138] 3389 (ms-wbt-server) : Connection timed out

(root@kali)~[/home/kali]
#
```

Figura 15. HTTPS al Web Client responde 200; SSH y RDP al Jump no conectan.

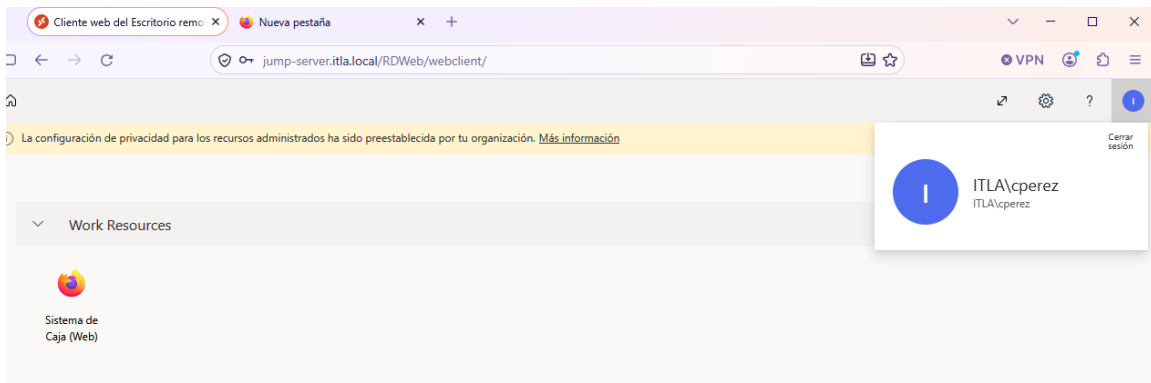


Figura 16. En el Web Client, cperez solo ve el Sistema de Caja.

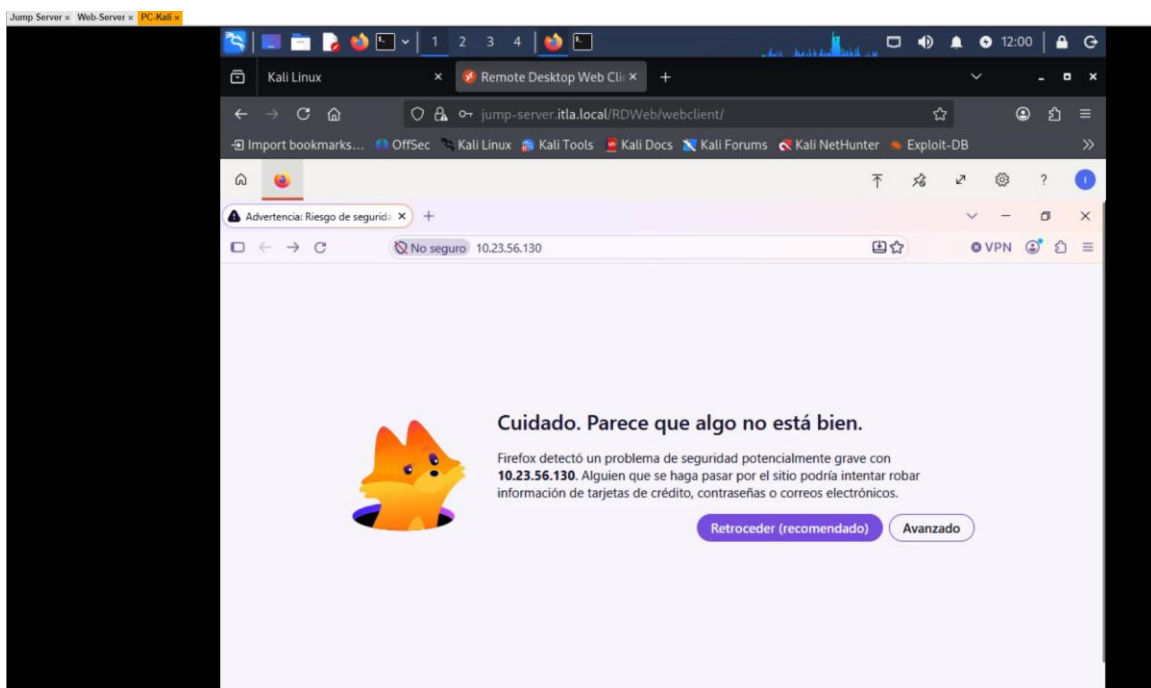
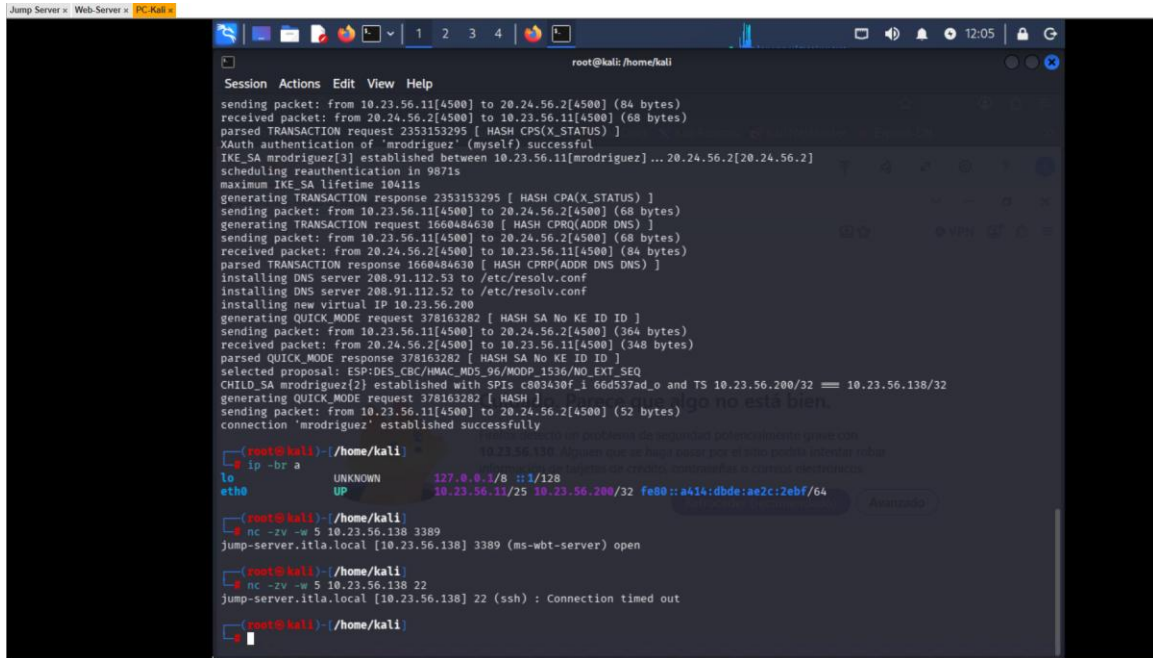


Figura 17. El Firefox del Jump Server llega al Web Server por HTTPS.

FortiGate VM64-KVM FG-Server								
X Action: Deny: policy-violation Add Filter								
	Date/Time	Source	Device	Destination	Application Name	Result	Policy	
	2026/10/08 13:06:34	cperez (10.23.56.200)		10.23.56.138		Deny: policy violation	Deny-SSH-Basico (3)	
	2026/10/08 13:06:34	cperez (10.23.56.200)		10.23.56.138		Deny: policy violation	Deny-SSH-Basico (3)	
	2026/10/08 13:06:33	cperez (10.23.56.200)		10.23.56.138		Deny: policy violation	Deny-SSH-Basico (3)	
	2026/10/08 13:06:31	cperez (10.23.56.200)		10.23.56.138		Deny: policy violation	Deny-SSH-Basico (3)	
	2026/10/08 09:37:05	10.23.56.138	JUMP-SERVER	148.103.0.32		Deny: policy violation	0	
	2026/10/08 09:36:53	10.23.56.138	JUMP-SERVER	148.103.13.34		Deny: policy violation	0	
	2026/10/08 09:36:47	10.23.56.138	JUMP-SERVER	148.103.13.34		Deny: policy violation	0	

Figura 18. Log del bloqueo con la política Deny-SSH-Basico.

5.2 Usuario con privilegios (mrodriguez)



```
root@kali: /home/kali
Session Actions Edit View Help
sending packet: from 10.23.56.11[4500] to 20.24.56.2[4500] (84 bytes)
received packet: from 20.24.56.2[4500] to 10.23.56.11[4500] (68 bytes)
parsed TRANSACTION request 2353153295 [ HASH CPS(X-STATUS) ]
XAuth authentication of 'mrodriguez' (myself) successful
IKE_SA mrodriguez[3] established between 10.23.56.11[mrodriguez] ... 20.24.56.2[20.24.56.2]
scheduling reauthentication in 9871s
maximum IKE_SA lifetime 10411s
generating TRANSACTION response 2353153295 [ HASH CPA(X-STATUS) ]
sending packet: from 10.23.56.11[4500] to 20.24.56.2[4500] (68 bytes)
generating TRANSACTION request 1660484630 [ HASH CPRQ(ADDR DNS) ]
sending packet: from 10.23.56.11[4500] to 20.24.56.2[4500] (68 bytes)
received packet: from 20.24.56.2[4500] to 10.23.56.11[4500] (84 bytes)
parsed TRANSACTION response 1660484630 [ HASH CPRP(ADDR DNS DNS) ]
installing DNS server 208.91.112.53 to /etc/resolv.conf
installing new virtual IP 10.23.56.200
generating QUICK_MODE request 378163282 [ HASH SA No KE ID ID ]
sending packet: from 10.23.56.11[4500] to 20.24.56.2[4500] (364 bytes)
received packet: from 20.24.56.2[4500] to 10.23.56.11[4500] (348 bytes)
parsed QUICK_MODE response 378163282 [ HASH SA No KE ID ID ]
selected proposal: ESP:DES_CBC/HMAC_SHA1_96/MD5/1836/NO_EXT_SEQ
CHILD_SA mrodriguez[2] established with SPIs c803430f_i 66d537ad_o and TS 10.23.56.200/32 == 10.23.56.138/32
generating QUICK_MODE request 378163282 [ HASH ]
sending packet: from 10.23.56.11[4500] to 20.24.56.2[4500] (52 bytes)
connection 'mrodriguez' established successfully

root@kali: /home/kali
# ip -br a
lo UNKNOWN 127.0.0.1/8 ::1/128
eth0 UP 10.23.56.11/25 10.23.56.200/32 fe80::a414:dbde:ae2c:2ebf/64

root@kali: /home/kali
# nc -zv -w 5 10.23.56.138 3389
jump-server.itla.local [10.23.56.138] 3389 (ms-wbt-server) open

root@kali: /home/kali
# nc -zv -w 5 10.23.56.138 22
jump-server.itla.local [10.23.56.138] 22 (ssh) : Connection timed out

root@kali: /home/kali
```

Figura 19. mrodriguez conecta la VPN; el RDP al Jump abre y el SSH no.

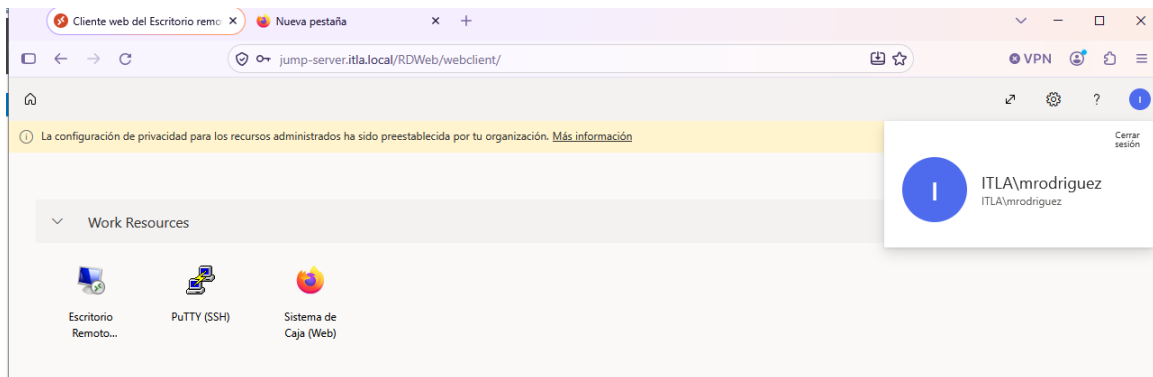


Figura 20. mrodriguez ve las tres RemoteApps.

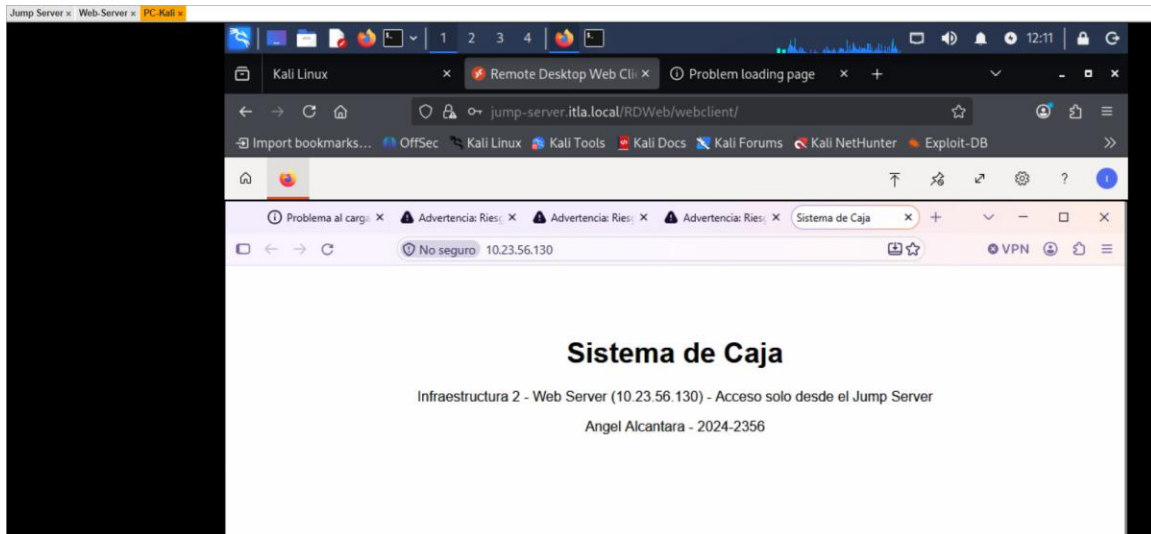


Figura 21. Sistema de Caja cargado desde el Jump Server.

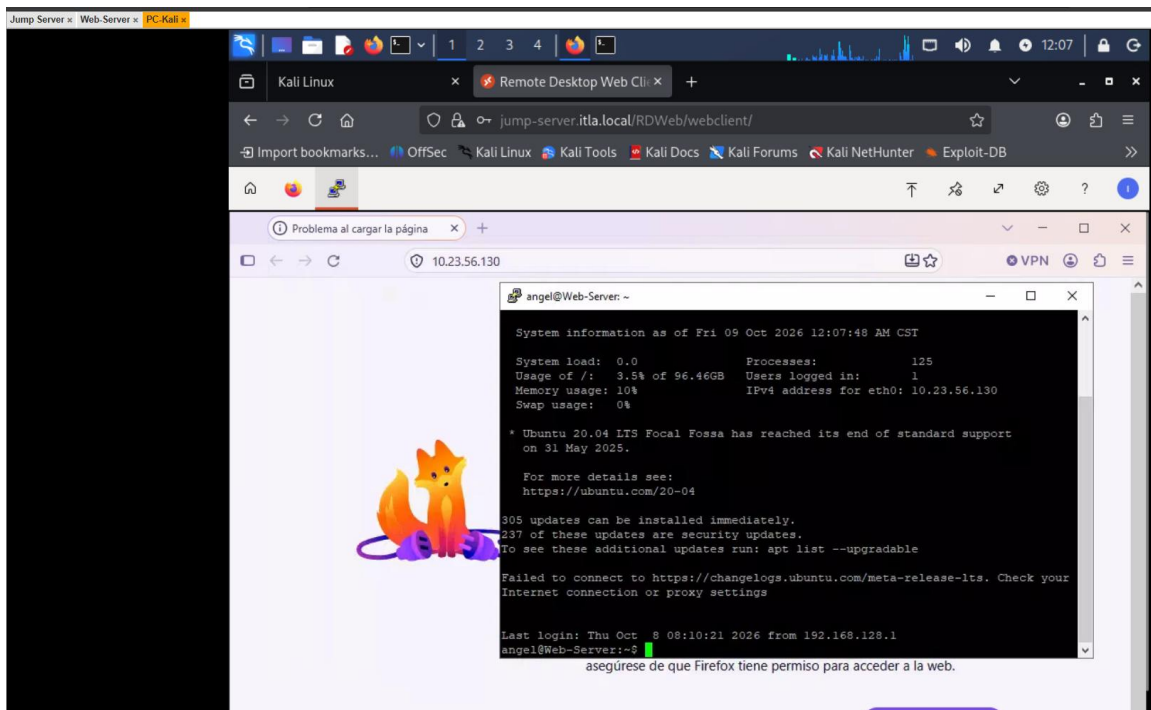


Figura 22. PuTTY entra al Web Server por SSH.

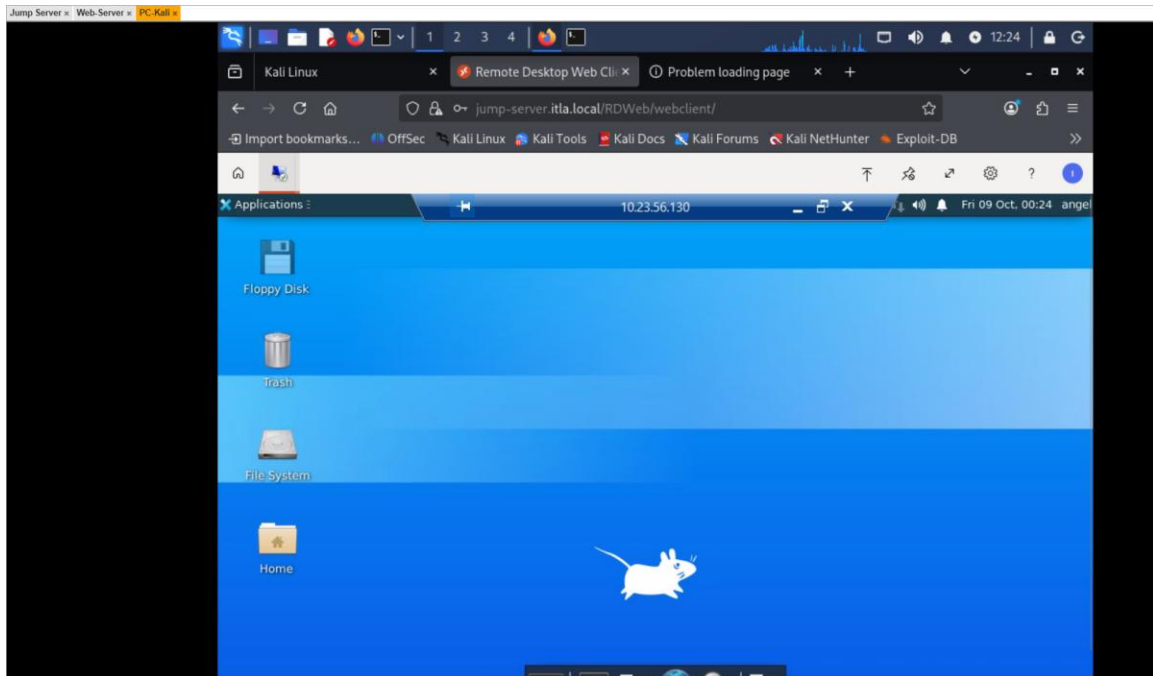


Figura 23. Escritorio Remoto al Web Server con escritorio XFCE.

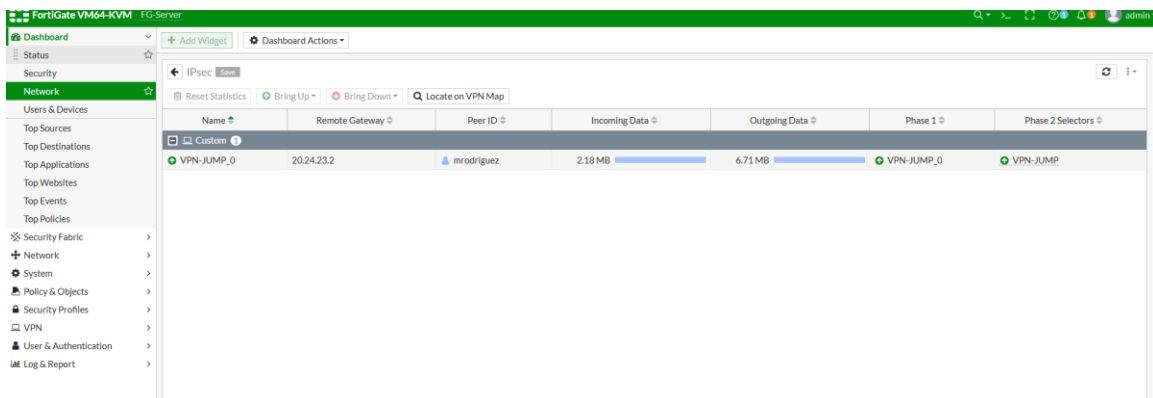


Figura 24. Túnel activo en Dashboard > Network > IPsec.

6. Conclusión

Con esta infraestructura el usuario nunca toca el Web Server directamente: entra por la VPN, solo puede llegar al Jump Server y desde ahí usa las aplicaciones que le tocan según su grupo. El usuario sin privilegios queda limitado a la web y su intento de SSH queda bloqueado y registrado, mientras que el usuario con privilegios puede administrar el servidor por SSH y RDP. Lo que más trabajo dio fue el RD Web Client, que necesitó el RD Gateway y ajustar el arranque del Connection Broker para funcionar bien.